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Test Booklet Series

TEST BOOKLET

C

GENERAL STUDIES (P) 2026 – Test – 6332

Time Allowed: Two Hours

Maximum Marks: 200

INSTRUCTIONS

1. IMMEDIATELY AFTER THE COMMENCEMENT OF THE EXAMINATION, YOU SHOULD CHECK THAT THIS BOOKLET DOES **NOT** HAVE ANY UNPRINTED OR TURN OR MISSING PAGES OR ITEMS, ETC. IF SO, GET IT REPLACED BY A COMPLETE TEST BOOKLET.
2. ENCODE CLEARLY THE TEST BOOKLET SERIES **A, B, C** OR **D** AS THE CASE MAY BE IN THE APPROPRIATE PLACE IN THE ANSWER SHEET.
3. You have to enter your Roll Number on the Test Booklet in the Box provided alongside. **Do NOT** write anything else on the Test Booklet.
4. This Test Booklet contains **100** items (Questions). Each item is printed in **English**. Each item comprises four responses (answers). You will select the response which you want to mark on the Answer Sheet. In case you feel that there is more than one correct response with you consider the best. In any case, choose **ONLY ONE** response for each item.
5. You have to mark all your responses **ONLY** on the separate Answer Sheet provided. See direction in the answers sheet.
6. All items carry equal marks. Attempt all items. Your total marks will depend only on the number of **correct responses** marked by you in the answer sheet. For **every incorrect** response **1/3rd of the allotted marks** will be deducted.
7. Before you proceed to mark in the Answer sheet the response to various items in the Test booklet, you have to fill in some particulars in the answer sheets as per instruction sent to you with your Admission Certificate.
8. After you have completed filling in all responses on the answer sheet and the examination has concluded, you should hand over to Invigilator only the answer sheet. You are permitted to take away with you the Test Booklet.
9. Sheet for rough work are appended in the Test Booklet at the end.

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE ASKED TO DO SO

1. With reference to human diseases, consider the following:

1. Haemophilia
2. Thalassemia
3. Sickle Cell Anaemia
4. Arthritis

How many of the diseases given above are genetic diseases?

- (a) Only one
- (b) Only two
- (c) Only three
- (d) All Four

2. Consider statements regarding the model of the atom:

1. Thomson's model explained that atoms are electrically neutral.
2. Rutherford proposed that there is a positively charged center in an atom called the nucleus.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

3. In the context of the recently developed model of the human embryo without eggs and sperm using stem cells (SCs), consider the following statements:

1. Adult Stem Cells (ASCs) are found in various tissues and organs, whereas, Embryonic Stem Cells (ESCs) are derived from the inner cell mass of a blastocyst following in vitro fertilization.
2. ASCs are Pluripotent, capable of differentiating into almost any cell type in the body, whereas ESCs are Multipotent cells that can differentiate into limited cell types.
3. ASCs are found during the early stages of embryonic development, whereas ESCs Present throughout one's lifetime.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

4. A concave lens is used for correction of which of the following eye defects?

1. Myopia
2. Hypermetropia
3. Presbyopia

Select the correct answer using the code given below.

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

5. A man is pushing hard against a rigid brick wall. In this context, which among the following is correct?

- (a) The internal energy used by the man is zero.
- (b) The work done by the man is zero.
- (c) The force applied by the man is zero.
- (d) None of the above

6. They are biodegradable, fire-resistant, and insulative bricks that combine husk, sawdust, and fungal spores to create a fibrous network that solidifies into a lightweight material. They reduce the building sector's carbon footprint and are also used as paneling materials in interior designs.

Which among the following is described in the above passage?

- (a) Ferrock
- (b) Hempcrete
- (c) Earthbag
- (d) Mycelium bricks

7. Which of the following statements correctly differentiate between saturated and unsaturated fatty acids?

- 1. Unsaturated fatty acids have one or more double bonds between their carbon atoms.
- 2. Saturated fatty acids have only single bonds between their carbon atoms.
- 3. Unsaturated fats, such as olive oil and fish oil, are typically liquid at room temperature.
- 4. Saturated fats, such as butter and ghee, are typically solid at room temperature.

Select the correct answer using the code given below.

- (a) 1, 2, 3 and 4
- (b) 1 only
- (c) 1 and 2 only
- (d) 1, 2 and 3 only

8. Consider the following statements with respect to Next Generation Launch Vehicle (NGLV):

- 1. The Next Generation Launch Vehicle will have a maximum payload capability of 30 tonnes to Low Earth Orbit.
- 2. The Next Generation Launch Vehicle will be designed as a fully reusable system.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

9. Consider the following statements:

- 1. Osmium is the densest metal.
- 2. Bromine is the only non-metal found in the liquid state under normal conditions.
- 3. Technetium was the first man-made element.
- 4. All the rare earth metals are from the lanthanide series in the periodic table.

Which of the statement given above are correct?

- (a) 2 and 4 only
- (b) 1 and 3 only
- (c) 1, 2 and 3 only
- (d) 1, 2, 3 and 4

10. With reference to human blood, consider the following statements:

- 1. Plasma constitutes more than half of the total blood volume.
- 2. Red blood cells contain haemoglobin, which is responsible for oxygen transport.
- 3. White blood cells are primarily involved in clotting of blood.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

11. An allotrope is each of the multiple forms in which an element can exist. In this context, which of the following is an allotrope of Carbon?

1. Diamond
2. Graphite
3. Graphene
4. Fullerenes

Select the correct answer using the code below.

- (a) 1 and 2 only
- (b) 1, 2, 3 and 4
- (c) 3 and 4 only
- (d) 2, 3 and 4 only

12. Consider the following:

1. Litmus
2. Red Cabbage
3. Onion
4. Turmeric

Which of the above can be used to indicate the presence of acid or base in a solution?

- (a) 1 and 2 only
- (b) 1, 2 and 3 only
- (c) 3 and 4 only
- (d) 1, 2, 3 and 4

13. With reference to the Indo-Pacific Oceans Initiative (IPOI), consider the following statements:

1. It was launched by India at the East Asia Summit in 2019.
2. It focuses on the Security and Growth for All in the Region (SAGAR) vision.
3. It is a legally binding framework for maritime cooperation in the Indo-Pacific region.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 1 and 2 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

14. Regarding the Priority Sector Lending Guidelines (PSL), 2025, consider the following:

1. Banks are barred from co-lending with NBFCs to meet PSL targets.
2. The PSL target for Small Finance Banks is 60% of their Adjusted Net Bank Credit.
3. Banks are permitted to treat export credit extended to the Agriculture sector as PSL loans.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 1 and 2 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

15. The archaeological site of Keeladi, recently in the news due to palaeoclimate reconstruction studies, is located at:

- (a) Semi-arid plateau region of Madhya Pradesh
- (b) Lower Ganga plains of West Bengal
- (c) Vaigai river basin in southern Tamil Nadu
- (d) Godavari river basin in Andhra Pradesh

16. Consider the following statements with regard to radiation:

1. Transfer of heat by radiation requires a medium.
2. All hot bodies radiate infrared radiation.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

17. What is the primary biological agent responsible for Root Wilt Disease in coconut palms?

- (a) Viruses
- (b) Nematodes
- (c) Fungi
- (d) Phytoplasmas

18. Which of the following glands is both exocrine and endocrine in function?

- (a) Thyroid
- (b) Adrenal
- (c) Pancreas
- (d) Pituitary

19. Consider the following statements regarding microorganisms:

1. All bacteria are pathogenic in nature.
2. Pathogens can include viruses, bacteria, fungi and protozoa.
3. Archaea are commonly associated with human infectious diseases.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

20. With reference to the 'Carbon Credit Trading Scheme (CCTS)' in India, consider the following statements:

1. It was notified under the Energy Conservation Act, 2001.
2. The Bureau of Energy Efficiency issues Carbon Credit Certificates (CCCs).
3. One Carbon Credit Certificate is equivalent to one tonne of CO₂ equivalent.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 1 and 3 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

21. Consider the following pairs:

Species	Key Characteristic / Status
---------	-----------------------------

- | | |
|-------------------------|--------------------------------------|
| 1. Ganges River Dolphin | : Relies on echolocation to navigate |
| 2. Indus River Dolphin | : Punjab's State Aquatic Animal |
| 3. Irrawaddy Dolphin | : Unique water-spitting behavior |

How many of the above pairs are correctly matched?

- (a) Only one pair
- (b) Only two pairs
- (c) All three pairs
- (d) None of the pairs

22. With reference to the immune system, what does the term "Innate Immunity" refer to?

- (a) It is an immunity acquired by an individual after exposure to a specific pathogen or through vaccination.
- (b) It refers to first line of defence that provides immediate, non-specific protection against Pathogens.
- (c) It is a type of immunity that involves antigen-specific responses mediated exclusively by antibodies.
- (d) It is a type of immunological memory that enables rapid response upon reinfection.

23. Consider the following pairs:

Places in the News	Located in / Associated with
--------------------	------------------------------

- | | |
|------------------------|-----------------|
| 1. Kamchatka Peninsula | : Pacific Ocean |
| 2. Diego Garcia | : Indian Ocean |
| 3. Negev Desert | : South America |

How many of the above pairs are correctly matched?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

24. Which of the following is used in a fluorescent lamp to produce visible light?
- Mercury vapour
 - Phosphor
 - Helium
 - Astatine
25. Recently seen in the news, the Bletchley Declaration is associated with:
- Promoting international trade
 - Encouraging nuclear disarmament
 - Providing guidelines for AI development and usage
 - Establishing a common currency
26. The 'Monroe Doctrine', recently seen in the news, is primarily related to:
- Assertion of U.S. primacy in the Western Hemisphere by opposing external interference.
 - Promotion of democratic institutions and electoral processes in Latin America.
 - A global digital currency framework led by the US Treasury.
 - A multilateral treaty for the conservation of the Amazon Basin.
27. Which among the following is the correct description of a redox reaction involving two molecules?
- Both molecule loses electrons
 - one loses electrons and is reduced, the other gains electrons
 - one loses electrons and is oxidized, the other gains electrons
 - one gains electrons and is oxidized, the other loses electrons
28. Which of the following best explains the excellent dissolving property of water?
- A pH of 7, which makes water neither acidic or basic
 - A density of roughly 1 gm/cm³ at room temperature
 - The polar nature of water molecules
 - None of the above
29. Consider the following statements regarding the status and designations of Kaziranga National Park (Assam):
- It is a designated National Park.
 - It is a recognized Tiger Reserve.
 - It is an Important Bird Area (IBA).
 - It is a UNESCO World Heritage Site.
- How many of the statements given above are correct?
- Only one
 - Only two
 - Only three
 - All four
30. Atomic species namely protium, deuterium and tritium are isotopes of which of the following elements?
- Carbon
 - Hydrogen
 - Helium
 - Nitrogen
31. Consider the following statements regarding biomolecules:
- Carbohydrates act as immediate as well as reserve sources of energy in living organisms.
 - All enzymes are globular proteins in nature.
 - Certain gut microorganisms synthesise vitamins essential for human metabolism.
- Which of the statements given above are correct?
- 1 and 2 only
 - 2 and 3 only
 - 1 and 3 only
 - 1, 2 and 3

32. With reference to Einstein rings, consider the following statements:

1. It is a ring of light around a dark matter formed via gravitational lensing.
2. Recently, the NASA James Webb space telescope discovered them for the first time.

Which among the above statements is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

33. The phenomenon of twinkling of stars is due to

- (a) atmospheric refraction of starlight
- (b) dispersion of light by atmosphere
- (c) diffraction of light of dust particles in the atmosphere
- (d) None of the above

34. With reference to the force of friction, consider the following statements:

1. It is the resistive force acting between bodies that tends to oppose and damp out motion.
2. It works only on moving objects.
3. Friction force is greater if a rough surface is involved than smooth surface.

Which of the statements given above is/are correct?

- (a) 1 and 2 only
- (b) 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

35. With reference to Alternating Current (AC) and Direct Current (DC), consider the following statements:

1. Both AC and DC flow in the direction of the flow of electrons.
2. In AC, the direction of the electric charge reverses periodically.
3. A battery produces only direct current.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 1 and 3 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

36. With reference to the small modular reactors (SMRs), consider the following statements:

1. They have a power generation capacity of up to 500MW per unit, which is half a traditional nuclear power reactor.
2. They can be pre-fabricated, shipped, and installed on the power generation site.
3. No external power is required to shut down an SMR.

How many of the above statements are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

37. With reference to the reactivity of metals, which of the following is the most reactive element?

- (a) Silver
- (b) Aluminium
- (c) Lead
- (d) Copper

38. Which of the following statements is/are correct regarding the sound waves?
1. Sound waves cannot travel through a vacuum.
 2. The particles oscillate up and down about their mean position as the wave travels.
 3. In a given medium under same physical conditions, the speed of sound changes drastically with a change in frequency.
- Select the correct answer using the code given below.
- (a) 1 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3
39. In a colloidal system, a liquid dispersed in a gas is known as:
- (a) Sol
(b) Emulsion
(c) Aerosol
(d) Foam
40. The term 'ECOFIX', recently seen in the news in the context of sustainable development, refers to:
- (a) A carbon-capture technology for thermal power plants
(b) An organic fertiliser produced from steel industry effluents
(c) A pothole-repair mix developed using waste-to-wealth principles
(d) A satellite-based system for monitoring rural road quality
41. Which of the following applications/phenomena are based on the Doppler effect?
1. Echolocation by bats.
 2. Sonic boom
 3. Police radar for measuring the speed of vehicles
- Select the correct answer using the code given below.
- (a) 1 and 2 only
(b) 1, 2 and 3
(c) 3 only
(d) 1 and 3 only

42. With reference to the Bagurumba Dance, consider the following statements:
1. It is a traditional folk dance of the Bodo community of Assam.
 2. It is traditionally performed only by men during harvest festivals.
- Which of the statements given above is/are correct?
- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2
43. A highly trained singer can hit a note that causes glass to shatter. Which of the following phenomenon exhibited by the sound waves is responsible for this?
- (a) Reverberation
(b) Boundary behaviour
(c) Resonance
(d) Interference
44. With reference to Tidal Locking, consider the following statements:
1. Tidal locking is a synchronization between the Moon's rotation and its orbit, caused by gravitational forces.
 2. The impact of solar winds on the Moon's rotation is the primary cause of tidal locking between celestial bodies.
 3. Tidal locking leads to the distortion of the Moon's shape and alternating high and low ocean tides.
- How many of the statements given above are correct?
- (a) Only one
(b) Only two
(c) All three
(d) None

45. With reference to important ports that have featured in recent geopolitical and connectivity initiatives, arrange the following ports from West to East:
- Port of Piraeus - Haifa Port - Jebel Ali Port - Chabahar Port
 - Haifa Port - Port of Piraeus - Jebel Ali Port - Chabahar Port
 - Port of Piraeus - Haifa Port - Chabahar Port - Jebel Ali Port
 - Haifa Port - Port of Piraeus - Chabahar Port - Jebel Ali Port
46. With reference to TSAT-1A Satellite, consider the following statements:
- It is India's first private earth observation satellite.
 - It was launched by ISRO.
 - It was sent into space as part of the Bandwagon-1 rideshare mission.
- How many of the statements given above are correct?
- Only one
 - Only two
 - All three
 - None
47. Capillary action is defined as the motion of liquids inside very narrow spaces without the assistance of, and most of the time in opposition to, external forces such as gravity. In this context, which of the following phenomena involve capillary action?
- A sponge absorbing water.
 - Burning of a candle.
 - Roots of plants drawing water from the soil.
- Select the correct answer using the code given below.
- 1 only
 - 2 only
 - 1 and 2 only
 - 1, 2 and 3

48. Consider the following pairs:
- | Region often mentioned in the news | Country |
|------------------------------------|-----------|
| 1. Ituri | : Ukraine |
| 2. Khartoum | : Sudan |
| 3. Chernihiv | : Spain |
- How many of the pairs given above are correctly matched?
- Only one
 - Only two
 - All three
 - None
49. A person stands on loose sand and then lies down on it. He goes deeper into the sand while standing than while lying down. Which one of the following is the reason?
- The weight of the body is lesser while standing up.
 - The density of the human body is lesser than sand
 - The pressure on the sand is higher while standing up
 - There is more friction between the body and the sand while lying down
50. Project Sanjay, recently seen in the news, is associated with:
- Indian Army's initiative to develop a medium-altitude, long-endurance unmanned aerial vehicle.
 - Indian Army's project to build a Battlefield Surveillance System for faster decision-making.
 - Indian Navy's project for indigenous construction of nuclear submarines.
 - ISRO's project for a new satellite launch vehicle.

- 51.** Consider the following statements regarding prokaryotic organisms:
1. Prokaryotes lack membrane-bound organelles.
 2. Their genetic material is not enclosed within a nuclear membrane.
 3. Some prokaryotes are obligate intracellular parasites.
- Which of the statements given above are correct?
- (a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3
- 52.** Bioluminescence, in which light is naturally produced by certain living organisms, involves the oxidation of which of the following?
- (a) Luciferin
(b) Bilirubin
(c) Melanin
(d) Lignin
- 53.** Which of the following is/are correct about the image of an object formed by a plane mirror?
1. The image formed is erect.
 2. It is at the same distance behind the mirror as the object is in front of it.
 3. It is laterally inverted with respect to the object.
- Select the correct answer using the code given below.
- (a) 1 and 2 only
(b) 2 and 3 only
(c) 1, 2 and 3
(d) 3 only

- 54.** With reference to the Atal Pension Yojana (APY), consider the following statements:
1. It is a Central Sector Scheme.
 2. Any citizen of India between the age of 18 and 60 years is eligible to join the scheme.
 3. It guarantees a minimum monthly pension after attaining the age of 60 years.
- Which of the statements given above is/are correct?
- (a) 1 only
(b) 2 only
(c) 1 and 3 only
(d) 1, 2 and 3
- 55.** Consider the following historical events:
1. The Delhi Conspiracy Case of 1912 involving an attack on Viceroy Lord Hardinge.
 2. The founding of the Indian Independence League in Tokyo in 1942.
 3. The organizational groundwork for the Azad Hind Fauj (INA).
- How many of the above events are associated with Rash Behari Bose?
- (a) Only one
(b) Only two
(c) All three
(d) None
- 56.** Consider the following statements:
1. Transcription involves the synthesis of RNA using DNA as a template.
 2. Transcription occurs in the nucleus in eukaryotic cells.
 3. RNA polymerase is required for transcription.
- Which of the statements given above are correct?
- (a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

57. Consider the following pairs:
- | Chemical reaction | Type |
|--|--------------------------|
| 1. Burning of coal | : Decomposition reaction |
| 2. Formation of water from oxygen and hydrogen | : Combination reaction |
| 3. Corrosion | : Oxidation reaction |
- Which of the pairs given above are correctly matched?
- (a) 1 and 2 only
(b) 2 and 3 only
(c) 1, 2 and 3
(d) 1 and 3 only
58. With reference to India, the terms 'Lambadi', 'Kurukh', and 'Mundari' pertain to:
- (a) Dance forms of North-East India
(b) Traditional councils
(c) Pre-historic cave paintings
(d) Tribal languages
59. Which of the following are base quantities as per the SI (International System of Units)?
1. Mass
 2. Length
 3. Charge
 4. Temperature
- Select the correct answer using the code given below.
- (a) 1, 2 and 3 only
(b) 1 and 2 only
(c) 1, 2 and 4 only
(d) 3 and 4 only
60. Consider the following diseases:
1. Tuberculosis
 2. Malaria
 3. Plague
- How many of the above diseases are caused by bacteria?
- (a) Only one
(b) Only two
(c) All three
(d) None

61. In the context of international relations, the term 'Diplomatic White Spaces' refers to:
- (a) International waters and polar regions where no single nation holds territorial sovereignty.
(b) Gaps in global leadership where coordination is required but no credible convenor exists.
(c) Neutral buffer zones established between warring nations under UN peacekeeping mandates.
(d) Secret diplomatic channels used for back-channel negotiations during bilateral trade disputes.
62. In Chemistry, a pH scale is used to measure the acidity or basicity of aqueous or other liquid solutions. In this context, arrange the following in decreasing order of pH.
1. Lemon Juice
 2. Milk of Magnesia
 3. Caustic Soda
- Select the correct answer using the code given below.
- (a) 3-1-2
(b) 3-2-1
(c) 2-1-3
(d) 1-2-3
63. Consider the following pairs:
- | Common Name | Chemical Formula |
|---------------------|---|
| 1. Plaster of Paris | : $\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$ |
| 2. Baking Soda | : Na_2CO_3 |
| 3. Washing Soda | : $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ |
| 4. Gypsum | : $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ |
- How many pairs given above are correctly matched?
- (a) Only one
(b) Only two
(c) Only three
(d) All four

- 64.** Consider the following statements regarding reproduction:
1. Asexual reproduction does not involve formation of gametes.
 2. Sexual reproduction generates genetic variation among offspring.
 3. All multicellular organisms reproduce sexually.
- How many of the statements given above are correct?
- (a) Only one
(b) Only two
(c) All three
(d) None
- 65.** With reference to LandSat Next, consider the following statements:
1. The satellite program will map the earth's changing surface to support climate monitoring and disaster management.
 2. It is a joint initiative of the Japan Aerospace Exploration Agency and the Australian government.
 3. It will be launched around 2030-2031 and consist of three identical satellites spanning the visible and infrared spectrum.
- Which of the statements given above are correct?
- (a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3
- 66.** Which of the following secreted the Insulin?
- (a) Alpha cells of pancreas
(b) Beta cells of pancreas
(c) Liver cells
(d) Adrenal cortex

- 67.** With reference to passive immunity, consider the following statements:
1. Passive immunity involves transfer of pre-formed antibodies from one individual to another.
 2. Passive immunity provides long-lasting protection.
- Which of the statements given above is/are correct?
- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2
- 68.** Consider the following statements with reference to transverse waves:
1. In transverse waves, the particles are displaced parallel to the direction the wave travels.
 2. Mechanical transverse waves cannot propagate in gaseous media.
 3. Secondary seismic waves are a type of transverse waves.
- Which of the statements given above is/ are correct?
- (a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3
- 69.** "Pauli engine" is sometimes talked about with reference to which one of the following?
- (a) Converting the energy difference between two quantum states of a group of atoms into work.
(b) Production of a new RNA based vaccine.
(c) Providing radio frequency based wireless communication for short distances.
(d) Mimicking the structure and functions of the human brain using the artificial neural networks.

70. In the context of gravitational force and electromagnetic force, consider the following statements:

1. While gravitational force is always attractive, electromagnetic force can be both attractive and repulsive.
2. While gravitational force is only dependent on mass, the electromagnetic force is dependent on both mass and charge.
3. While gravitational force is dependent upon the distance between the two bodies, the electromagnetic force is independent of it.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

71. With reference to the AMoRE Experiment, consider the following statements:

1. It is an international collaboration to search for neutrinoless double-beta decay in molybdate crystal.
2. It has operated since 2015 at Fermi National Accelerator Laboratory in the United States.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

72. Consider the following:

Diseases	Factors
-----------------	----------------

- | | |
|---------------|------------|
| 1. Filariasis | : Virus |
| 2. Dengue | : Protozoa |
| 3. Chickenpox | : Virus |
| 4. Malaria | : Bacteria |

How many of the pairs given above are correctly matched?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

73. Consider the following statements with reference to electric fuse:

1. The principle behind the operation of the fuse is the heating effect of the electrical current.
2. It is connected in parallel with the electric component.

Which one of the statements above the fuse is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

74. Which of the following statements best explains the reason for airplane windows being rounded rather than square?

- (a) Rounded windows have higher tensile strength as compared to squared windows.
- (b) Stress is more equally distributed along the edges in the rounded windows as compared to squared windows.
- (c) Rounded windows have lesser surface area and thus are less expensive.
- (d) Rounded windows are easier to fit as compared as squared windows.

75. With reference to electrolysis, consider the following statements:

1. Electrolysis is a process in which an electric current is passed through a liquid or solution to bring about a chemical change.
2. In the electrolysis of water, oxygen gas is released at the cathode, and hydrogen gas is released at the anode.
3. Electrolysis is used in electroplating metals and purification of metals.

Which of the statements given above is/are correct?

- (a) 1 and 3 only
- (b) 2 and 3 only
- (c) 1 only
- (d) 1, 2 and 3

76. The terms 'Mistral', 'Gemma' and 'Granite', which were recently seen in the news, are:
- (a) Small language models used for specialised AI tasks
 - (b) Missile defence systems used for aerial interception
 - (c) Tropical weather systems formed over warm oceans
 - (d) Deep-sea exploration vehicles for underwater missions

77. With reference to Hypersonic weapons, consider the following statements:

1. A hypersonic weapon travels faster than Mach 3 or three times the speed of sound.
2. Hypersonic weapons can bypass most traditional missile defense systems.
3. The Avangard is a nuclear-capable, hypersonic missile developed by the United States of America.

How many of the above statements are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

78. Consider the following statements about the cell wall:

1. It provides mechanical strength and protection to plant cells.
2. It is a non-living structure present outside the plasma membrane.
3. It controls the selective permeability of substances into the cell.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

79. The area known as the 'Nakhchivan Autonomous Republic' sometimes appears in the news in the context of events related to:

- (a) South-East Asia
- (b) Caucasus region
- (c) Middle East region
- (d) Eastern Europe

80. With reference to the ROMAN Space Telescope, consider the following statements:

1. It is an ESA infrared space observatory that probes dark energy and exoplanets in the universe.
2. Set to launch in 2026, it will be situated at Lagrange point 2.
3. It will be a successor observatory of WFIRST, launched in 2010.

How many statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

81. Which one of the following is the correct description of "Near Net Shape Forming"?

- (a) It is a metal-forming process that creates components in their final shape without the need for any post-processing or machining.
- (b) It is an advanced manufacturing method that produces components with close-to-final shape with minimal waste and machining.
- (c) It is a method of shaping ceramics at high temperature to directly produce final mechanical parts with zero shrinkage.
- (d) It is a design approach in additive manufacturing where parts are made in layers, which allows for direct digital-to-physical shape transfer.

82. Consider the following:
Assertion (A): As compared to ionic compounds, covalent compounds have low melting and boiling point.
Reason (R): Inter-molecular force in covalently bonded molecules is stronger than Intra-molecular force.
- (a) A and R both are true, and R is the correct explanation for A
(b) A and R both are true, and R is the NOT the correct explanation for A
(c) A is correct, R is incorrect
(d) A is incorrect, R is correct

83. While it is stressful to watch a movie on TV in a dark room, it is not while watching a movie in a traditional theatre. Consider the following reasons:
1. In a traditional theatre, we receive analog pictures while on TV we receive images in pixels.
 2. In the theatre, the view is almost completely occupied by the screen unlike that in the case of TV.
- Which of the statements given above is/are correct?
- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

84. Consider the following schemes:
1. Atal Innovation Mission
 2. GENESIS
 3. NIDHI
 4. ASPIRE
- How many of the above schemes aim to promote the startup ecosystem in India?
- (a) Only one
(b) Only two
(c) Only three
(d) All four

85. A body is falling freely under the action of gravity alone in a vacuum. Which one of the following is correct?
- (a) In the beginning, the kinetic energy of the body is zero.
(b) As the body falls, the potential energy of the body keeps increasing.
(c) The sum of potential energy and kinetic energy is different at all points.
(d) All of the given statements are correct.

86. With reference to the Global Positioning System (GPS), consider the following statements:
1. It's a navigation system based on radio signals.
 2. It operates using trilateration principles.
 3. It is a network of 24 satellites.
- How many of the statements given above are correct?
- (a) Only one
(b) Only two
(c) All three
(d) None

87. Consider the following statements regarding *Aqua Regia*:
1. It is a mixture of concentrated hydrochloric acid and concentrated nitric acid in the ratio of 1:3.
 2. It is one of the few reagents that are able to dissolve gold and platinum.
- Which of the statements given above is/are **not** correct?
- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

88. With reference to the Nobel Peace Prize, consider the following statements:
1. The Peace Prize is awarded by the Norwegian Nobel Committee.
 2. The title and honour associated with the prize are transferable with prior approval.
- Which of the statements given above is/are correct?
- (a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2
89. Which plant hormone is known as the “stress hormone” in plants?
- (a) Auxin
(b) Gibberellin
(c) Cytokinin
(d) Absciscic acid
90. Consider the following statements with reference to Heavy water and Hard water:
1. Though the molecular composition of both Heavy water and Hard water are same, they differ in their mineral composition.
 2. Heavy water is radioactive and Hard water is not.
 3. Only Hard water is used as a moderator in nuclear reactors.
- Which of the statements given above is/are correct?
- (a) 1 only
(b) 2 and 3 only
(c) 1 and 2 only
(d) None

91. Consider the following pairs:
- | <i>Type of immunity</i> | <i>Example</i> |
|--------------------------|----------------|
| 1. Physiological barrier | : Skin |
| 2. Cellular defence | : Macrophages |
| 3. Chemical mediator | : Antibodies |
- Which of the pairs given above are correctly matched?
- (a) 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1 and 2 only
92. Which of the following is/are a chemical process?
1. Melting of an ice cube
 2. Formation of vinegar from sugarcane juice
 3. Mixing sugar in water
- Select the correct answer using the code given below.
- (a) 1 only
(b) 2 and 3 only
(c) 2 only
(d) 1, 2 and 3
93. The term “Willow” recently in the news relates to which among the following?
- (a) IKEA's new completely recyclable furniture lines
(b) Russia's new hypersonic missile system
(c) Tesla's upcoming space exploration mission
(d) Google's new quantum computing chips
94. Consider the following enzymes:
1. Pepsin
 2. Trypsin
 3. Rennin
- Which of the above is/are protein-digestion enzymes in humans?
- (a) 1 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

95. With reference to various tokamaks, consider the following pairs:

Countries	Name
-----------	------

- | | |
|-------------------|--------------|
| 1. China | : EAST |
| 2. European Union | : JET |
| 3. Japan | : JT-60 |
| 4. France | : Tore Supra |

How many of the above pairs are correctly matched?

- (a) Only one pair
- (b) Only two pairs
- (c) Only three pairs
- (d) All four pairs

96. With reference to the genome sequencing project, consider the following statements:

- 1. The IndiGen program aims to collect 10,000 genetic samples from citizens across India to build a reference genome.
- 2. The Genome India Project aims to undertake whole genome sequencing of thousands Indian individuals representing diverse ethnic groups from India.
- 3. The Human Genome Project aims to determine the DNA sequence of the entire human genome.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

97. Which of the following phenomena of light waves are involved in the formation of a rainbow?

- 1. Refraction
- 2. Total internal reflection
- 3. Dispersion

Select the correct answer using code given below.

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

98. With reference to enzymes, consider the following statements:

- 1. Enzymes reduce the activation energy of reactions.
- 2. Enzymes are consumed during biochemical reactions.
- 3. Enzyme activity is influenced by temperature and pH.

How many of the statements given above are correct?

- (a) Only one
- (b) Only two
- (c) All three
- (d) None

99. With reference to the Sukanya Samriddhi Yojana (SSY), consider the following statements:

- 1. It was launched as part of the Beti Bachao, Beti Padhao campaign.
- 2. Only one Sukanya Samriddhi Account is permitted per girl child.
- 3. The SSY account matures after 21 years, before which no withdrawal is permitted under any circumstances.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 1 and 2 only
- (c) 2 and 3 only
- (d) 1, 2 and 3

100. Which one of the following countries has its own Space Station System?

- (a) China
- (b) Canada
- (c) Israel
- (d) Japan